DESIGN AND DEVELOPMENT OF A WEB-BASED HOTEL AND RESTAURANT MANAGEMENT SYSTEM FOR HOSPITALITY MANAGEMENT STUDENTS OF THE POLYTECHNIC UNIVERSITY OF THE PHILIPPINES - SAN JUAN CAMPUS

System Analysis & Design Project
Presented to the Faculty of the Diploma in Information Technology
Polytechnic University of the Philippines
San Juan City

In Partial Fulfillment of the Requirements for the Diploma in Information Technology

By

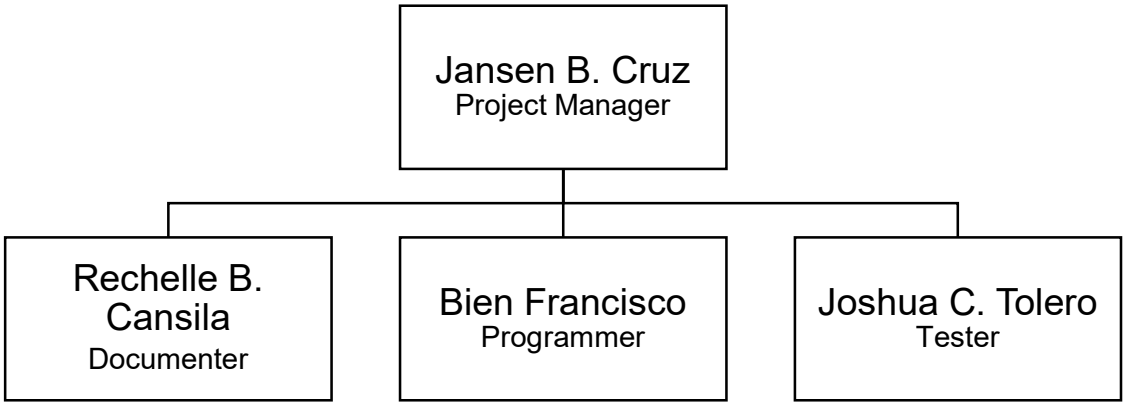
Cansila, Rechelle B.
Cruz, Jansen B.
Francisco, Bien
Tolero, Joshua C.

2026

Project Team

The project team is composed of project manager, programmer, documenter, and tester. The following organizational chart shows the roles of the members.

Figure 1. Organizational Chart



Chapter 1

INTRODUCTION

PROJECT CONTEXT

The hospitality industry increasingly relies on digital technologies to support hotel and restaurant operations. Computerized management systems help establishments manage reservations, guest registration, check-in and check-out, housekeeping, payment and billing, restaurant transactions, and guest services. As these operations become more technology-driven, Hospitality Management students need opportunities to develop practical skills in using computerized systems relevant to the industry.

The increasing role of technology in hospitality highlights the need to strengthen digital learning in hospitality education. Busulwa et al. (2024) examined the integration of digital capabilities in hospitality and tourism curricula and found that digital transformation capabilities were not consistently integrated across programs and subjects. Similarly, Stylianou and Pericleous (2025) found that hospitality and tourism graduates had basic digital skills but limited practical experience with industry-specific hotel software. These findings emphasize the importance of providing students with opportunities to gain hands-on experience with technologies used in hospitality operations.

Digital experiential learning can help address this need. He and Zhang (2025) found that digital experiential learning improved students' ability to use digital technology and positively influenced their learning. Likewise, Masset and Weisskopf (2025) found that simulation activities in hospitality education increased student

engagement and supported improved learning outcomes. These studies highlight the potential of technology-based learning environments to help students apply hospitality concepts and develop practical skills through simulated activities.

In response to this need, this study proposes the Design and Development of a Web-Based Hotel and Restaurant Management System for Hospitality Management Students of the Polytechnic University of the Philippines – San Juan Campus. The system will provide a school-based environment where students can practice realistic hotel and restaurant operations using simulated data and transactions. Hotel functions will include online and front-desk reservations, guest registration, check-in and check-out, room management, housekeeping, payment and billing, night audit, and amenities or guest services. Restaurant functions will include relevant restaurant transactions, table management, and related services.

The proposed system is expected to provide Hospitality Management students with an accessible and easy-to-use environment for practicing interconnected hotel and restaurant operations through a computerized system similar to those used in actual establishments. It may also support the Hospitality Management program by providing a platform for practical activities and technology-based learning. Therefore, this study will focus on developing a realistic and ease of use web-based system that allows students to practice hotel and restaurant management operations within the school.

STATEMENT OF THE PROBLEM

The study seeks to address the need for a web-based hotel and restaurant management system that can provide Hospitality Management students of the Polytechnic University of the Philippines – San Juan Campus

with an accessible environment for practicing realistic and interconnected hospitality operations.

Specifically, the study seeks to address the following problems:

Main Problem

Limited opportunities for Hospitality Management students to practice integrated hotel and restaurant operations using a computerized management system that reflects actual hospitality workflows.

Other Problems

1. Limited opportunities to practice hotel reservation processes through online and front-desk reservation channels, along with guest registration, check-in and check-out, room management, housekeeping, payment and billing, and other hotel services.

2. Limited opportunities to practice restaurant operations, including restaurant transactions, table management, and related guest services using a computerized system.

3. The need for an accessible and easy-to-use web-based system that enables students to practice hotel and restaurant management activities effectively.

OBJECTIVE OF THE STUDY

The study aims to design and develop a web-based hotel and restaurant management system for Hospitality Management students of the Polytechnic University of the Philippines – San Juan Campus that provides a realistic,

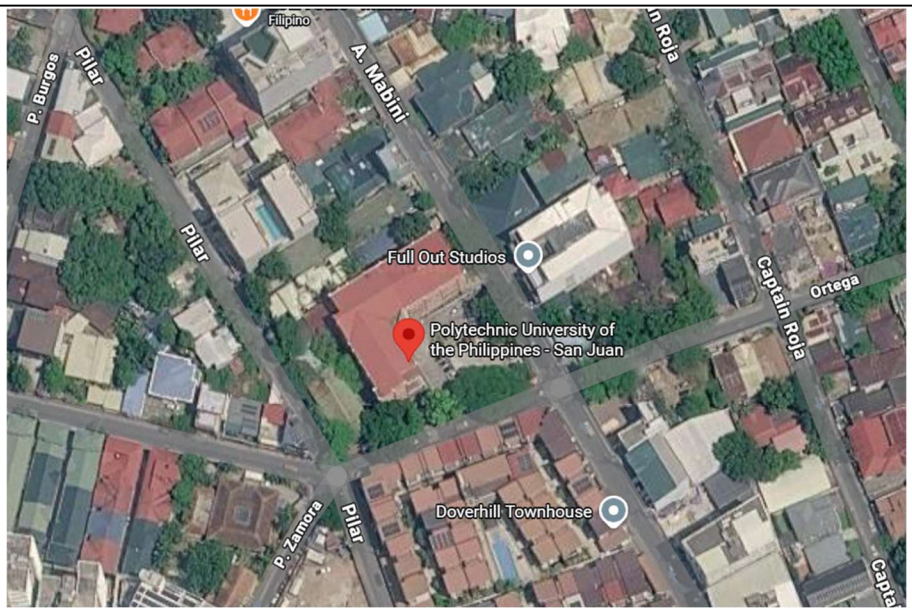
accessible, and easy-to-use environment for practicing interconnected hotel and restaurant operations.

Specifically, the study aims to:

1. Design and develop a web-based hotel and restaurant management system that provides Hospitality Management students with a realistic environment for practicing computerized hospitality operations.
2. Integrate essential hotel operations, including online and front-desk reservations, guest registration, check-in and check-out, room management, housekeeping, payment and billing, night audit, and amenities or guest services.
3. Integrate relevant restaurant operations, including restaurant transactions, table management, and related guest services, into the system.
4. Develop an accessible and easy-to-use interface that enables Hospitality Management students to navigate and perform hotel and restaurant management activities effectively.

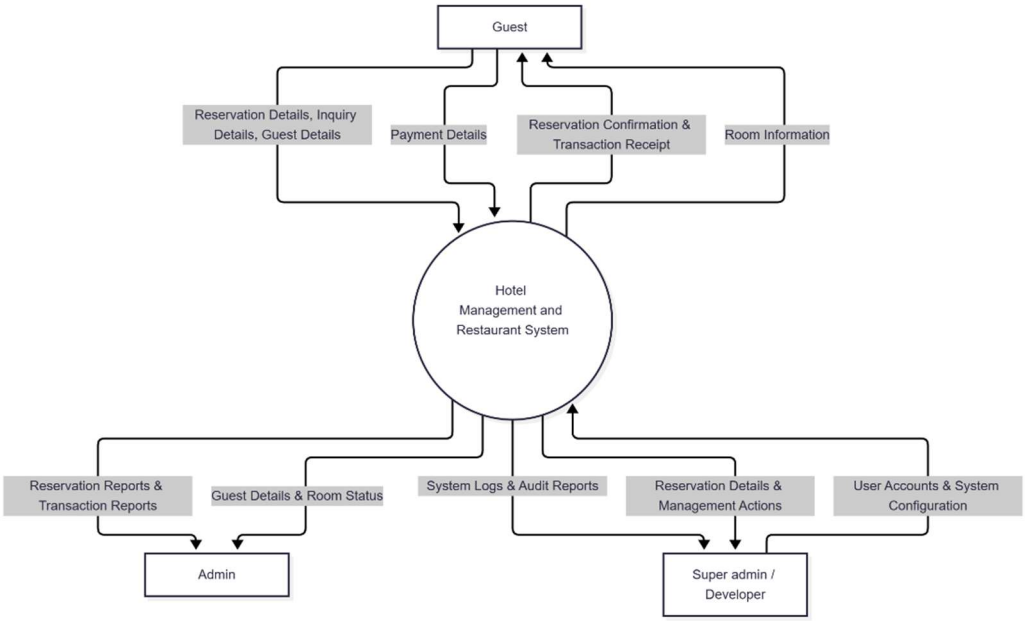
SETTINGS OF THE STUDY

Figure 1. Location of the Polytechnic University of the Philippines – San Juan Campus



The study is will take place at the Polytechnic University of the Philippines – San Juan City Campus 223 Ortega, Brgy. Additon Hills, San Juan City, Metro Manila, with the Hospitality Management Department as the client and the primary users are Bachelor of Science in Hospitality Management (BSHM) students, taking courses on front office, lodging, room division and food and beverage operations, and the system administrator. Instructors take part as evaluators and have no dedicated system role. The system will be used in the campus laboratory and simulation room, with the kiosk/tablet placed in a simulated front-desk area. All guest, payment and identity data are simulated or sample data; no real guest or financial information is processed.

CONTEXT DIAGRAM



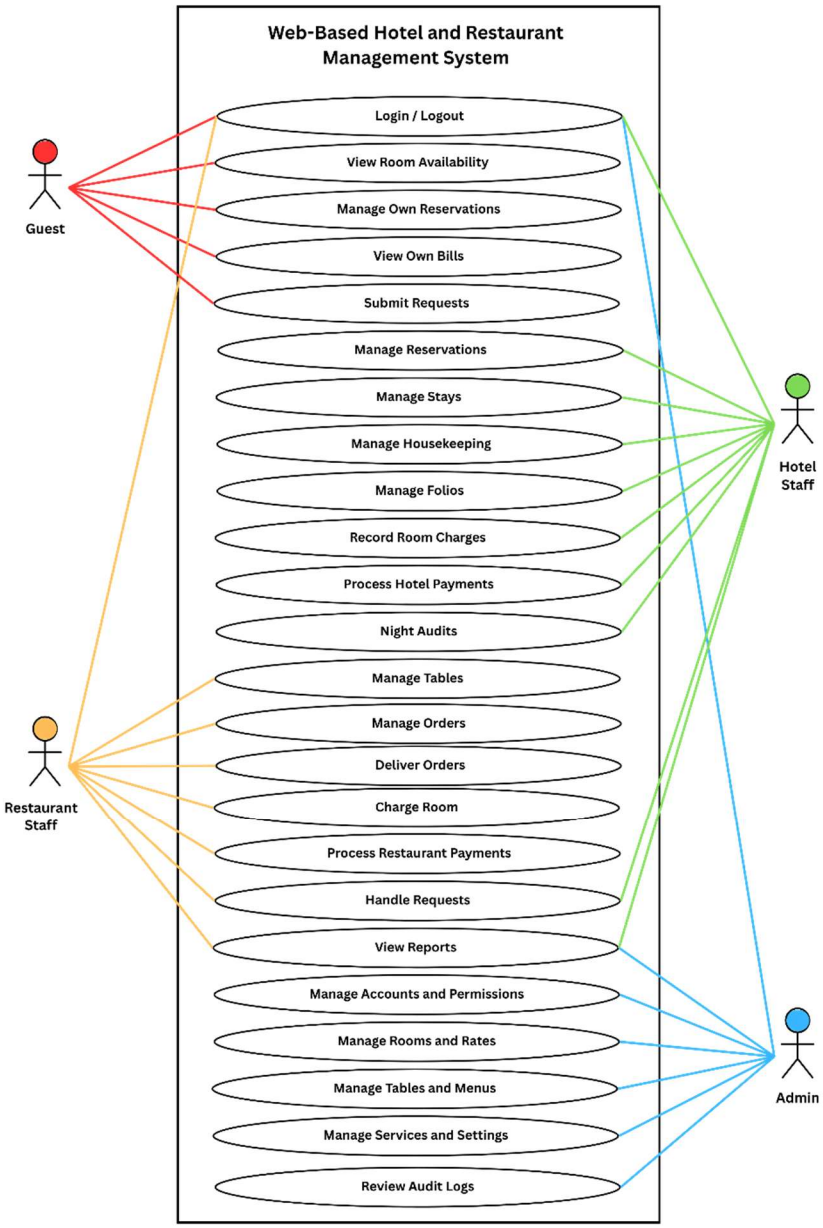
The Context Diagram (Data Flow Diagram Level 0) illustrates the high-level boundaries and data exchanges between the core system—the Hotel Management and Restaurant System—and its primary external entities. This architectural overview demonstrates how different users interact with the platform without exposing internal database structures or low-level program code.

The primary external entities and their respective data exchanges include:

- Guest: Interacts with the system to submit Reservation Details, Inquiry Details, and Guest Details for booking rooms or restaurant tables. The guest also provides Payment Details and receives Room Information, Reservation Confirmations, and Transaction Receipts in return.
- Admin/Staff: Serves as the operational manager for daily facility workflows. This entity reviews Reservation Reports & Transaction Reports for business tracking, and accesses Guest Details & Room Status to monitor front-office and room assignments.

- Super Admin / Developer: Handles system-level configuration and maintenance oversight. This entity manages User Accounts & System Configuration, processes administrative Reservation Details & Management Actions, and reviews System Logs & Audit Reports to maintain platform security and data integrity.

USE CASE DIAGRAM

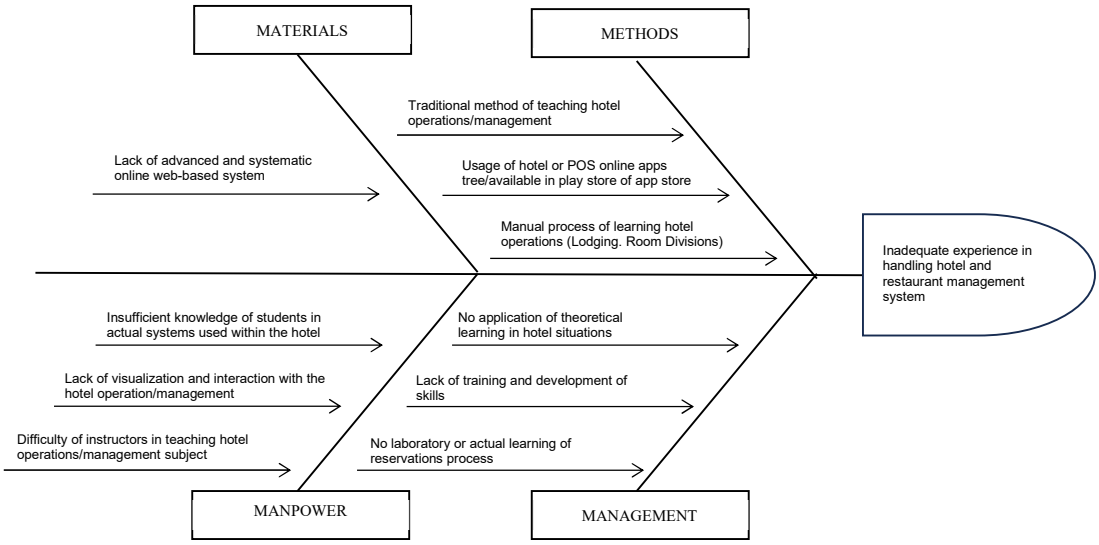


The use case diagram illustrates the interactions of four actors—Guest, Hotel Staff, Restaurant Staff, and Administrator—with the proposed Web-Based Hotel and Restaurant Management System. Guests can view room availability, create, modify, or cancel their own reservations, view their bills, and submit requests or concerns.

Hotel Staff can manage reservations and stays, including guest registration, room assignment, check-in, and checkout. Their functions also cover housekeeping, guest folios, room charges, hotel payments, and night audit. Restaurant Staff can manage table availability and orders, record room-service deliveries, process restaurant payments, and post eligible restaurant charges to a checked-in guest's folio. Both staff groups can handle assigned requests and view permitted operational reports.

The Administrator manages user accounts and permissions, rooms, room types, rates, tables, menus, services, and system settings. The Administrator can also view reports and review audit logs. All four actors have login and logout access. Guest access is restricted to their own records, while staff functions depend on assigned permissions. The diagram represents one integrated system with separate hotel and restaurant workspaces linked through shared guest and billing records.

FISHBONE DIAGRAM



The fishbone's head contains the primary effect of the absence of technology, such as the Online Hotel Management System at the Polytechnic University of the Philippines San Juan, which is the inadequate experience of BSHM students in handling hotel management systems. The contributors that lead to the main issue include that the materials are lacking or there is no existing advanced and systematic online web-based system that the HM Students can utilize. The traditional method of teaching hotel operations/management is generally used, and the students learn through a manual process and by using online applications available in the play store. Another source of the issue is the people, where the students have insufficient knowledge of existing systems used within the hotel, there is a lack of visualization and interaction regarding the hotel The fishbone's head contains the primary effect of the absence of technology, such as the Online Hotel Management System at the Polytechnic University of the Philippines San Juan, which is the inadequate experience of BSHM students in handling hotel management systems.

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GANTT CHART

Activity Name	August				September				October				November				December			
1. Project Manager																				
2. Analysis and Design																				
3. Documentation																				
4. Development of the project																				
5. Testing the project																				
6. Project Presentation																				
7. Revisons																				
8. Completion																				

11. Hardware Requirements for Development

The following specifications are planning targets for a small development and campus demonstration setup. They are not measured production capacity requirements. Available laboratory equipment will be assessed before final deployment.

Component	Minimum planning target	Preferred target
Core	64-bit dual-core processor; 8 GB RAM	Recent quad-core processor; 16 GB RAM
Storage	256 GB SSD with at least 20 GB free for project, dependencies, and test data	512 GB SSD with space for versioned backups
Display and input	1366 × 768 display, keyboard, and mouse	1920 × 1080 display for design and testing
Local demonstration server	Development computer running Apache, PHP, and MySQL or MariaDB	Dedicated laboratory computer for concurrent exercises
Client computers	Browser-capable computer connected to the server	Desktop or laptop with stable wired or wireless connection

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